

CHEMISTRY 101 · WEEK 1 QUIZ

Chemistry 101 — Week 1

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. Which is the best description of an atom?

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- A. The smallest building block that everything around you is made of.
- B. A small bubble of gas that floats around in the air we breathe.
- C. A tiny living creature that can be seen only with a microscope.
- D. The smallest piece of a solid before it melts into a liquid.

2. In the formula H_2O , what does the small 2 mean?

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- A. Each water molecule contains two oxygen atoms.
- B. It takes two water molecules to make one drop of water.
- C. Hydrogen is the second element on the list of elements.
- D. Each water molecule contains two hydrogen atoms.

3. How many atoms in total are in one molecule of carbon dioxide, CO_2 ?

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- A. 4
- B. 2
- C. 1
- D. 3

4. About how much of the air you breathe is oxygen?

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- A. Only a trace (less than 1%)
- B. About half (50%)
- C. About one-fifth (21%)
- D. Almost all of it (about 95%)

5. Most of the air is a gas your body mostly ignores – breathes in, breathes right back out. Which gas?

- ☐ A. Hydrogen
- ☐ B. Nitrogen
- ☐ C. Carbon dioxide
- ☐ D. Oxygen

6. In the lungs, which way do the two gases move?

- ☐ A. Oxygen goes from blood into air; carbon dioxide goes from air into blood.
- ☐ B. Both gases move from air into blood, and the body stores them.
- ☐ C. Both gases move from blood into air, and the body replaces them.
- ☐ D. Oxygen goes from air into blood; carbon dioxide goes from blood into air.

7. Why does the body have to get rid of carbon dioxide?

- ☐ A. It is heavier than oxygen, so it would sink and collect in the feet.
- ☐ B. It takes up space that the body needs for storing water.
- ☐ C. Too much of it makes the blood acidic, which upsets the body's chemistry.
- ☐ D. It turns blood a darker color that the pulse oximeter cannot read.

8. In 1774, Joseph Priestley found that a mouse lived much longer in a jar of his newly discovered gas than in a jar of ordinary air. What did this show?

- ☐ A. The new gas (oxygen) was the part of air that living things actually need.
- ☐ B. The mouse had grown stronger from being kept inside a sealed jar.
- ☐ C. Ordinary air becomes poisonous to small animals in closed spaces.
- ☐ D. The new gas was lighter than air, so the mouse could breathe it more easily.

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)